

# Assessing and Training Young Children in Same and Different Relations Using the Relational Evaluation Procedure (REP)

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**Abstract** Previous research suggests the relational evaluation procedure (REP) is a useful means by which to assess and train relational responding. Most work so far has been with adults; however, given the potential utility of the REP for assessing and training relational responding, researchers need to investigate its use with young children. The current series of studies presents relevant data. Study 1 shows correlations between performance on a simple REP-based multi-level protocol (the NSD-REP) and cognitive and linguistic ability in a relatively large ( $n = 26$ ) sample of typically developing children (aged 2–5). Studies 2–4 involved training REP performance to criterion in a number of these children, using a multiple baseline design across participants in each case to show experimental control. These findings supplement previous data suggesting the utility of the REP for training relational responding in young children.

**Keywords** Relational frame theory (RFT) · Relational evaluation procedure (REP) · Children · Assessment · Training · Multiple baseline design

Many species can be trained to show nonarbitrary relational responding, which is based on physical characteristics of the stimuli related; for example, picking something physically bigger or smaller than a sample stimulus. However, according to relational frame theory (RFT; Hayes, Barnes-Holmes & Roche, 2001), language-able humans in addition show derived or arbitrarily applicable relational responding (AKA relational framing), which is based on aspects of the context ('contextual cues') that specify the relation independent of the physical characteristics of the stimuli. For example, if told that 'X is bigger than Y', I can derive that 'Y is smaller than X', without having to rely on properties of X and Y themselves but guided instead solely by the cue specifying the relationship (i.e., 'bigger').

RFT argues that we learn to engage in both nonarbitrary and arbitrarily applicable relational responding as a result of exposure to contingencies of reinforcement operating in the human verbal community. For example, in the case of nonarbitrary comparative relations, contingencies provided by the community will favor picking the physically larger stimulus if told to choose the bigger one of two and the physically smaller stimulus if told to choose the smaller one. Nonarbitrary relational responding in turn provides an important foundation for the emergence of arbitrarily applicable relational responding. After multiple exemplars of feedback for engaging in nonarbitrary comparative relational responding, for instance, eventually comparative relational responding might be emitted in contexts in which physical properties are no longer present or relevant and only contextual cues are available as antecedent guides for the response, such as in the 'X bigger than Y' example given above.

Comparative relations are only one variety of relation. Multiple other types have been demonstrated also, at both nonarbitrary and arbitrarily applicable levels, including same (e.g., '1 + 1 = 2'), different (e.g., 'Apples are different from oranges'), opposite (e.g., 'Day is opposite to night'), spatial (e.g., 'The TV is on the table'), temporal (e.g., 'Spring is before summer'),

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hierarchical (e.g., 'Potatoes are a type of vegetable'), etc. (Barnes-Holmes, Barnes-Holmes, Smeets, Strand & Friman, 2004; Dymond & Barnes, 1995; Dymond, Ng, & Whelan, 2013; Luciano, Gomez, & Rodriguez, 2007; May, Stewart, Baez, Freegard, & Dymond, under submission; O'Hora, Barnes-Holmes, & Stewart, 2014; O'Hora, Roche, Barnes-Holmes, & Smeets, 2002; Slattery & Stewart, 2014; Steele & Hayes, 1991). In addition, at the arbitrarily applicable level, all forms involve derivation-based properties including mutual entailment (e.g., if A is related to B, then B is related to A; for example, if  $A > B$  then  $B < A$ ), combinatorial entailment (e.g., if A is related to B and B is related to C, then A is related to C and vice versa; for example, if  $A > B$  and  $B > C$ , then  $A > C$  and  $C < A$ ) and transformation of function (i.e., if A has a function and is related to B, then the functions of B may be transformed based on the type of relation and type of function; for example, if A has approach functions and is opposite to B, then B may acquire avoidance functions).

From the RFT point of view, the variety and generativity of arbitrarily applicable relational responding are what underlie or facilitate the full complexity of human language and cognition (see Stewart, 2016 for a brief overview of evidence for this contention). For example, one recent series of studies, involving both typically developing children and children with developmental delay, has demonstrated correlations between performance on a protocol centered on arbitrarily applicable equivalence (sameness) relational responding and on standardised measures of language and cognition (e.g., Moran, Stewart, McElwee, & Ming, 2010; Moran, Stewart, McElwee, and Ming, 2014). As such, from the RFT perspective, it is critical to study the development of arbitrarily applicable relational responding from its roots in non-arbitrary relations, through its emergence via multiple exemplar training to its fullest expression in complex language. One methodology that RFT researchers have developed explicitly for this purpose is the relational evaluation procedure (REP; see Barnes-Holmes, Hayes, Dymond, & O'Hora, 2001; Stewart, Barnes-Holmes, & Roche, 2004). The REP is a methodology developed as a means of assessing and training relational framing under controlled laboratory conditions. The core feature of the REP is that it allows experimental participants to evaluate, or report on, the stimulus relation or relations that are presented on a given trial (e.g., Stewart et al., 2004). As an example of a typical REP type study, in Stewart et al. (2004), pre-training first established particular relational functions (i.e., 'same' and 'different') for arbitrary stimuli as well as establishing the functions of 'yes' and 'no' for two further arbitrary stimuli. Then, during an assessment phase, stimuli in a particular relation with each other (e.g., two same- or differently-coloured objects), a contextual cue (e.g., one of the two arbitrary shapes established as 'same' and 'different'), and the cues for 'Yes' and 'No' were presented. As predicted, participants chose the 'yes' comparison when the contextual cue correctly corresponded to the relationship between the coloured objects

(e.g., 'same colour' or 'different colour') and chose the 'no' comparison when it did not.

Once the functions of the contextual cues were established in Stewart et al., it became possible to generate multiple relatively complex patterns of relational responding (relational networks), each of which involved a totally novel set of stimuli, quickly and efficiently. This particular study was focused on modelling analogical reasoning. In previous work on analogy, the matching-to-sample (MTS) procedure had been used to generate analogy-like behavior, namely, the derivation of relations between relations. To do so in these cases, however, first extensive baseline training had been needed for each example of analogy. However, in Stewart et al., once the functions of the contextual cues had been established, it was possible to generate a novel analogical relational network based on completely novel stimuli on every trial. In this way, the REP as used in Stewart et al. and similar studies has arguably enabled researchers to explore patterns of relational framing in humans more effectively than older methodologies such as the MTS protocol could. In addition, more recently, it has also inspired the development of additional protocols offering further particular advantages for assessing and training relational framing in particular contexts using both adults and children as participants.

One example is the SMART (strengthening mental abilities with relational training) online protocol. The SMART protocol is essentially a systematically organised set of REP trials focused on the thoroughgoing assessment and training of core forms of relational framing. An early prototype of this online method was employed by Cassidy, Roche and Hayes (2011) to train both educationally typical and educationally disadvantaged children in core patterns of relational framing. The basic format involves the presentation of a relational network and a question as to the nature of the relation between two members of that network that could be answered by pressing either 'yes' or 'no' at the bottom of the screen. Results showed that months-long training at increasing levels of complexity in a number of frames including same, opposite, more and less resulted in substantial WISC IQ gains in both educationally normal (study 1) and educationally delayed children (study 2).

Another concept inspired by the REP is the implicit relational assessment protocol (IRAP), which has now become extremely useful and popular in the exploration of participants' verbal repertoires. The IRAP requires participants to respond quickly and accurately via response options such as 'yes/no' or 'true/false' to networks of relations. IRAP research has typically been focused on exploring attitudes (e.g., to race) in typically developed adult humans. However, it has also been employed in work focused on other subjects of psychological research such as intellectual or academic potential and has involved participants both younger and older than this. One recent domain of research, for example, has begun to focus on IRAP testing and training in typically developing children as well as children with educational and behavioural difficulties (e.g.,

Kilroe, Murphy, Barnes-Holmes, & Barnes-Holmes, 2014; Scanlon, McEnteggart, Barnes-Holmes, & Barnes-Holmes, 2014). For example, Kilroe et al. (2014) used the IRAP to explore relational responding in children with autism, while Scanlon et al. (2014) used it to explore gender bias and self-esteem in typically developing children as well as children with attention-deficit/hyperactivity disorder (ADHD) and dyslexia.

The previous paragraphs suggest both the importance of RFT as an empirically based approach to investigating human language and cognition, and of the REP as an effective methodology for pursuing basic and applied RFT research. As suggested, there are now a significant number of studies that have used the REP or its derivatives. Despite this, however, and despite the potential efficacy of the REP, in one arena, relatively little work has as yet been conducted. This is research examining the potential of this method as an assessment and training tool with very young children. As we have reviewed, the REP, in various forms, has been employed with children in a number of studies. However, in all such studies thus far, the children involved have tended to be of at least school age. Thus, as yet, no published work has examined the use of the REP with very young children. However, given the potential of the REP as an assessment and training tool, and the importance of efficiently assessing and training relational responding in young children, such work should arguably be a priority.

The aim of the current study was to begin to address this deficit in the literature. This study explored the use of an REP-based protocol referred to here as the nonarbitrary same different REP or NSD-REP as a potential tool for assessment and training in young typically developing preschool children from 2 to 5 years old. Children were first assessed for a repertoire of yes/no responding per se. The purpose of this test was to see whether children had a related though dissimilar repertoire to that targeted by the REP and thus might be in an approximately suitable range for assessment and training of the latter skillset. If successful, they were then exposed to the NSD-REP along with a series of other assessments. The NSD-REP involved three levels. Level 1 tested non-arbitrary same and different relations by requiring participants to press a button for either 'Same' or 'Different' in response to a pair of identical or non-identical pictures. Level 2 tested evaluation of non-arbitrary same and different by requiring participants to press a button for either 'yes' or 'no' in response to a pair of identical or non-identical pictures presented together with an auditory recording of the word 'same' or 'different'. Level 3 was similar to the latter level but increased task complexity by including the contextual cue 'not'. Participants were scored between 0 and 60 based on their performance (see Procedure). The purpose of scoring them in this way was not to develop a standardised scoring criterion but simply to facilitate comparison between performance on this protocol and that on other measures in the context of the current study. In experiment 1, performance on the NSD-REP was correlated

with performance on the other measures while in the remaining experiments, participants failing at one or other of the levels of the NSD-REP were trained to meet criterion on that level. While correlations between relational responding and performance on standardised measures of language and cognition in young children have been investigated previously, the current study represents an advance by employing an REP as the testing procedure. Furthermore, as explained, the investigation of the REP as a training procedure for relational responding in young children in studies 2–4 is also a departure, as little or no work has been conducted in this domain.

## Study 1

### Method

#### *Participants and Setting*

Prior ethical approval for this study was obtained from the Research Ethics Committee at the National University of Ireland Galway. Participants were twenty six children (14f), aged between 26 and 59 months ( $M = 42.8$ ;  $SD = 9.6$ ) recruited from three separate facilities in County Limerick in Ireland. Two were day care centers in Limerick city that catered for children from birth to five years, while the third was a Montessori school in County Limerick that catered for children aged 3 to 5. In the case of one of the two day care centers and the Montessori school, participants were included in the study if their parents had signed and returned consent forms to the class teacher, they had no special educational needs, as reported by the teacher/parent and they assented to participation. All participants attending these centers at the time of data collection were included on this basis. In the case of the second day care center, children from there were recruited primarily for the purposes of providing participants for experiment 4 and thus only children aged between 2 and 3 were recruited. Other than that, the same inclusion criteria as for the other two centers were also applied in this case.

#### *Materials*

##### **Assessment of Textual Responding To Contextual Cue**

**Words** This was a brief test of whether children could read the contextual cue words that would appear on the buttons of the NSD-REP (e.g., 'yes', 'no'). This test involved printed out versions of these words.

**Yes/No Prescreening Assessment** This was adapted from Shillingsburg, Kelley, Roane, Kissamore and Brown (2009). It involved 10 picture cards, each showing a single stimulus (e.g., ball, car, duck, bear). Each of the ten picture cards was presented in turn and in each case, the participant was asked a pre-devised question concerning the identity of the pictured

item to which either a 'yes' or a 'no' response would be appropriate (five required a 'yes' response and five a 'no' response; see [Appendix 1](#)). The questions were presented in a randomised order based on a random number generator.

**Stanford Binet Intelligence Scales 5th Edition Abbreviated Battery IQ (SB5-ABIQ)** The SB5 (Roid, 2003) is an individually administered test of intelligence and cognitive abilities for individuals between 2 and 85 years of age. The Abbreviated Battery IQ (ABIQ) scale of the SB5 used in this experiment consists of two subtests: one designated nonverbal (object series/matrices) and one designated verbal (vocabulary). The 'nonverbal' subtest 'is designed to assess fluid reasoning skills and includes sequential reasoning and classic matrix items wherein the person being tested is required to select the best alternative to complete a series or a matrix. The verbal subtest is designed to measure declarative knowledge and in particular vocabulary. Early items require a pointing response (e.g., item 1; 'Touch your mouth'). Mid-level items require single word responses (e.g., item 12; 'What's happening in this picture?' [accompanied by a picture of a person running]). Upper level items require that words be defined clearly (e.g., item 20; "What is a parrot?"). The SB5-ABIQ takes between 15 and 20 minutes to complete. For the purposes of the current research, total score was based on raw scores; hence, the 'combined scale' score for the SB5-ABIQ that we report in this study, which we refer to as the SB5 total (SB5 T) score, is simply the addition of the scores for each of the two subscales. The SB5-ABIQ has excellent internal consistency with an average coefficient of .91.

**Preschool Language Scales - 4th Edition (PLS-4)** This is a 130-item standardized measure of language skills in children from birth to 6 years 11 months (Zimmermann, Steiner & Pond, 2002). It takes 25–40 minutes to administer. It is a useful diagnostic and research tool that can be used to identify current comprehension and expressive language skills and can measure changes in language skills over time. The test is individually administered and includes tasks that assess skills in the areas of preverbal behaviors, as well as linguistic skills in the areas of semantics, morphology, syntax, integrative language skills, and preliteracy skills. It consists of two subscales that assess auditory comprehension (AC) and expressive communication (EC). The AC subscale (62 items) assesses skills such as attention to speakers, object play, comprehension of basic vocabulary, response to grammatical markers, identification of rhyming words and ability to make comparisons. The EC subscale (68 items) assesses skills including object naming, object description, expression of quantity and the use of grammatical markers. The PLS-4 yields norm-referenced test scores (standard scores, percentile ranks, and an age equivalent) for both subscales, as well as for total language (TL) score. Test-retest reliability has been reported as ranging between 0.82 and

0.95 for the subscale (AC and EC) scores and between 0.90 and 0.97 for the total (TL) scale score, while internal reliability has been reported as ranging from 0.66 to 0.96.

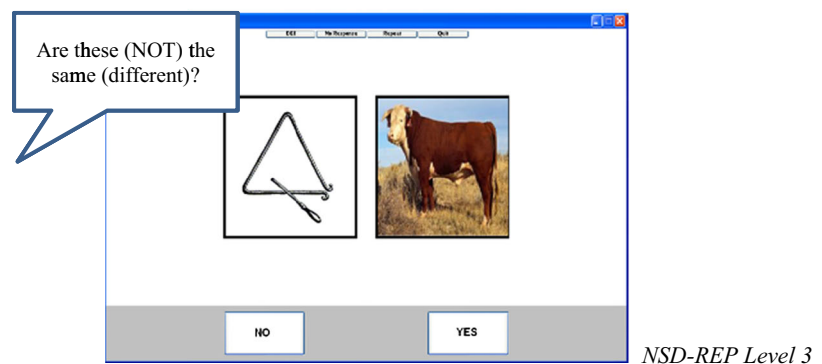
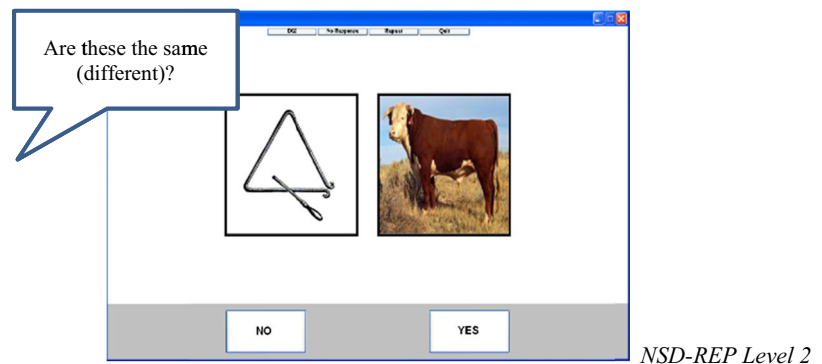
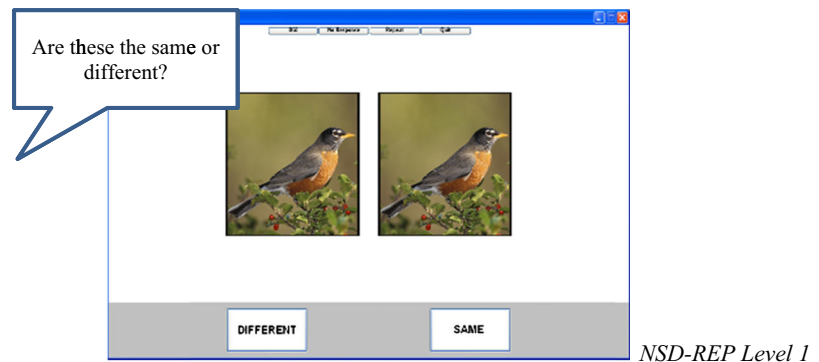
**Non-arbitrary Same Different Relational Evaluation Protocol (NSD-REP)** This was programmed in Livecode™ and was presented on a Dell Latitude 2100 touchscreen laptop computer. It consisted of three levels as follows: (1) selection of either the textual stimulus 'same' or 'different' given the auditory prompt "Are these the same or different?" and a pair of either identical or non-identical pictures; (2) selection of either the textual stimulus 'yes' or 'no' given either the auditory prompt 'Are these same?' or 'Are these different?' and a pair of either identical or non-identical pictures; and (3) selection of either the textual stimulus 'yes' or 'no' given either the auditory prompt 'Are these not the same?' or 'Are these not different?' and a pair of either identical or non-identical pictures (see Fig. 1). Each level involved 20 trials without feedback, on each of which a respondent could be scored as either 'correct' or 'incorrect'. A 'correct' response was defined as touching the correct onscreen stimulus while refraining from touching an incorrect stimulus, while an 'incorrect' response was defined as touching an incorrect stimulus or touching both correct and incorrect stimuli. The raw score per level was from zero to 20. As a means of controlling for chance responding, the raw score on each level was converted into a final score by subtracting ten and, if a negative score resulted, converting that into zero. This gave a range for a level final score of 0–10. A score of 8 or higher on this index was deemed a pass and allowed access to the next level. Level final scores were added to give a total score on the protocol.

### *Procedure*

**Yes/No Pre-Screening Assessment** All assessments were conducted at a table in a corner of the main activity room used by the day care facility. The first session involved administration of the yes/no assessment, the purpose of which was to ensure that each participant had the capacity to respond to simple yes and no questions. For this, the child sat opposite the experimenter. This assessment was conducted in one sitting and took approximately 2–4 minutes to complete. No feedback was provided during testing and the criterion was set at 90 %. All participants attained this criterion and proceeded to the next phase of assessment.

**Assessment of Textual Responding to Contextual Cue Words** Before further assessment, each participant was presented with a brief test of whether they could read the contextual cue words that would appear on the buttons of the NSD-REP (e.g., 'yes', 'no'). This test involved presenting them with printed versions of these words and asking them in each case whether they could say what word was being presented.

**Fig. 1** Illustrations of each of the three levels appearing in the NSD-REP



None of the children appeared to be able to read any of the words thus presented.

**Assessment with PLS-4 and SB5-ABIQ** Participants were subsequently assessed on both the PLS-4 and SB-5-ABIQ. The PLS-4 has two separate subtests focused on receptive and expressive communication respectively. In the case of each child, assessment using this measure was completed in two sessions, each lasting about 16 minutes. In a third session for each child, the SB5-ABIQ was administered, and took about 20 minutes to complete.

**NSD-REP Assessment** After PLS-4 and SB5-ABIQ testing was completed, children were exposed to the NSD-REP protocol. This assessed the evaluation of non-arbitrary same and different relations at three levels of complexity including (i) same and different (ii) yes and no and (iii) yes and no with not.

On each trial of Level 1 ('same and different'), two pictures (either identical or non-identical) were presented on the screen accompanied by an auditory prompt (i.e., 'Are these the same or different?'). In response, participants were required to press the 'same' button (which always appeared on the lower left of the screen) if the pictures were identical or the 'different'



button (which always appeared on the lower right of the screen) if not. No feedback was provided after a response; instead, the screen cleared and half a second later the next trial was presented. A total of 20 trials was presented. In level 2 ('yes and no'), two pictures (either identical or non-identical) were presented on the screen accompanied by one of two different auditory prompts (i.e., either 'Are these the same?' or 'Are these different?'). This time, participants were required to press either the 'yes' button (which always appeared on the lower left of the screen) or the 'no' button (which always appeared on the lower right of the screen). Again, no feedback was provided after a response; instead the screen cleared and half a second later the next trial was presented. A total of 20 trials was presented. Level 3 was identical in all respects to level 2, except that the auditory prompts were either 'Are these not the same?' or 'Are these not different?'.

On each of the three levels, responding during a trial was scored as either 'correct' or 'incorrect'. A 'correct' response was defined as touching the correct onscreen stimulus while refraining from touching an incorrect stimulus. An 'incorrect' response was defined as touching an incorrect stimulus or touching both correct and incorrect stimuli. All levels had the same criterion for number of correct responses needed to pass (i.e.,  $18/20 = 90\%$  of the raw score, which translates to  $8/10 = 80\%$  of the adjusted score). If a participant succeeded in meeting this criterion, they were deemed as passing that level and were then exposed to the next level. If a participant did not attain this target they were deemed to have failed that level.

In the case of all children, the NSD-REP assessment was completed in one session. The average time to complete the protocol was 5 mins 45 secs (one and a half minutes each for levels 1 and 2 and three minutes 45 seconds for level 3).

## Results and Discussion

### Scoring Protocols

Demographic information and scores for the NSD-REP, PLS-4 and SB5-ABIQ assessments are provided in Table 1. Scores for the NSD-REP were calculated as outlined in the Materials section. Scores for the PLS-4 (raw and age equivalent) and the SB5-ABIQ (verbal and non-verbal), were calculated as per their respective manuals (Zimmermann et al., 2002; Roid, 2003).

### Correlations Amongst the Measures

Table 2 displays a correlation matrix of Spearman rank order correlations among the experimental measures and some of their key subscales. This shows a correlation between the NSD-REP and PLS-4 total scores ( $\rho = 0.87$ ;  $p < 0.001$ ) as well

as between the former and both auditory comprehension ( $\rho = 0.88$ ;  $p < 0.001$ ) and expressive communication ( $\rho = 0.84$ ;  $p < 0.001$ ) PLS-4 subscales. The NSD-REP was also correlated with the SB5-ABIQ Total (SB5 ABIQ T) score ( $\rho = 0.82$ ;  $p < 0.001$ ) as well as with the scores for its two subscales;  $\rho = 0.79$  ( $p < 0.001$ ) with the verbal scale and  $\rho = .63$  ( $p = 0.0012$ ) with the non-verbal scale. As such, a key pattern of results was that the NSD-REP correlated with the standardised measures of both language (i.e., the PLS-4) and cognitive ability (i.e., the SB5-ABIQ).

Table 2 also reveals correlations amongst the standard measures of language and cognitive ability themselves. The SB5-ABIQ T scale was highly correlated with the PLS-4 and its subscales; it showed a correlation of 0.83 ( $p < 0.001$ ) with the total scale, of 0.87 ( $p < 0.001$ ) with auditory comprehension and 0.77 ( $p < .001$ ) with expressive communication. The SB5-ABIQ verbal subscale was also highly correlated with the PLS-4 and its subscales. It showed a correlation of 0.8 ( $p < 0.001$ ) with the total scale, of 0.83 ( $p < 0.001$ ) with auditory comprehension and of 0.75 ( $p < 0.001$ ) with expressive communication. Finally, the SB5-ABIQ non-verbal subscale also correlated with the PLS-4 and its subscales. In this case, the correlations were 0.67 ( $p < 0.001$ ) with the PLS-4 total scale, 0.72 ( $p < 0.001$ ) with auditory comprehension and 0.56 ( $p < 0.001$ ) with expressive communication.

When examining the correlations amongst the three main measures used, namely, the NSD-REP, the PLS-4 and the SB5-ABIQ, the highest correlation was that between the NSD-REP and the PLS-4 ( $\rho = 0.87$ ); meanwhile, the NSD-REP and the PLS-4 both correlated at a lower level but still strongly and comparably with the SB5-ABIQ (i.e.,  $\rho = 0.82$  and  $\rho = 0.83$ , respectively). The close correlation of the NSD-REP with both other measures used reflects the RFT conception that contextually controlled relational responding is the core process involved in language and cognition. The fact that the NSD-REP was correlated slightly more strongly with the dedicated language measure than the measure of cognition may reflect a closer connection between relational responding and focused language tasks than between relational responding and a broader range of tasks. Indeed, the pattern of correlations between the NSD-REP and the SB5-ABIQ subscales supports this suggestion; the correlation with the verbal subscale ( $\rho = 0.79$ ) was considerably higher than with the nonverbal subscale ( $\rho = 0.63$ ). Interesting also are the correlations between the NSD-REP and the PLS-4 subscales. In this case, it can be seen that the NSD-REP correlated more highly with the auditory comprehension subscale than with the expressive communication subscale and indeed it correlated more highly with the former than with the PLS-4 itself. This might be explained as being a result of the fact that the NSD-REP involves auditory as well as visual stimuli as part of the trials on all three stages and that touchscreen rather than spoken responses are required. Of course many of the

**Table 1** Demographic information and scores for initial NSD-REP, PLS-4 and SB5-ABIQ assessment for all participants in studies 1–4

| Pt. No. | Age  | Sex | Study | NSD-REP | PLS4 T | PLS4 AC | PLS4 EC | PLS4 AE | SB5 T | SB5 NV | SB5 V | SB5 AE | SB5-AB IQ |
|---------|------|-----|-------|---------|--------|---------|---------|---------|-------|--------|-------|--------|-----------|
| 1       | 4.00 | F   | 1,3   | 8       | 112    | 58      | 54      | 4.05    | 30    | 10     | 20    | 4.09   | 109       |
| 2       | 3.06 | F   | 1,3   | 9       | 113    | 54      | 59      | 4.06    | 25    | 8      | 17    | 3.11   | 103       |
| 3       | 3.00 | M   | 1,2   | 1       | 92     | 44      | 48      | 3.03    | 22    | 6      | 16    | 3.04   | 103       |
| 4       | 3.07 | M   | 1,2,3 | 5       | 79     | 40      | 39      | 2.07    | 23    | 9      | 14    | 3.07   | 100       |
| 5       | 3.10 | F   | 1     | 4       | 103    | 50      | 53      | 3.10    | 27    | 9      | 18    | 4.03   | 106       |
| 6       | 3.06 | M   | 1,2   | 5       | 94     | 47      | 47      | 3.04    | 24    | 9      | 15    | 3.09   | 103       |
| 7       | 3.05 | M   | 1,2,3 | 1       | 83     | 40      | 43      | 2.09    | 24    | 9      | 15    | 3.09   | 106       |
| 8       | 2.04 | M   | 1     | 0       | 65     | 27      | 38      | 2.00    | 11    | 3      | 8     | <2.00  | 88        |
| 9       | 2.01 | F   | 1     | 0       | 68     | 33      | 35      | 2.02    | 15    | 4      | 11    | 2.04   | 109       |
| 10      | 3.00 | F   | 1     | 0       | 86     | 41      | 45      | 2.11    | 23    | 10     | 13    | 3.07   | 109       |
| 11      | 2.06 | F   | 1     | 0       | 81     | 40      | 41      | 2.00    | 11    | 2      | 9     | <2.00  | 85        |
| 12      | 4.11 | M   | 1,3   | 8       | 109    | 52      | 57      | 4.03    | 18    | 8      | 10    | 4.03   | 91        |
| 13      | 4.06 | M   | 1     | 9       | 109    | 54      | 55      | 4.03    | 31    | 10     | 21    | 4.11   | 103       |
| 14      | 4.08 | F   | 1,3   | 10      | 110    | 56      | 54      | 4.03    | 32    | 10     | 22    | 5.01   | 103       |
| 15      | 4.05 | M   | 1,3   | 10      | 111    | 54      | 57      | 4.04    | 28    | 9      | 19    | 4.05   | 100       |
| 16      | 3.07 | M   | 1     | 0       | 86     | 40      | 46      | 2.11    | 22    | 9      | 13    | 3.04   | 103       |
| 17      | 4.01 | F   | 1     | 7       | 103    | 49      | 54      | 3.10    | 26    | 7      | 19    | 4.01   | 100       |
| 18      | 4.02 | F   | 1     | 9       | 109    | 50      | 59      | 4.03    | 30    | 8      | 22    | 4.09   | 103       |
| 19      | 3.07 | M   | 1     | 7       | 96     | 45      | 51      | 3.05    | 21    | 9      | 12    | 3.03   | 94        |
| 20      | 3.06 | M   | 1     | 0       | 80     | 39      | 41      | 2.08    | 13    | 5      | 8     | 2.01   | 73        |
| 21      | 4.09 | F   | 1,3   | 10      | 112    | 56      | 56      | 4.05    | 37    | 16     | 21    | 5.11   | 118       |
| 22      | 3.11 | M   | 1     | 10      | 95     | 46      | 49      | 3.04    | 24    | 9      | 15    | 3.09   | 100       |
| 23      | 2.08 | F   | 1,4   | 0       | 69     | 31      | 38      | 2.02    | 16    | 6      | 10    | 2.05   | 97        |
| 24      | 2.10 | F   | 1     | 0       | 68     | 30      | 38      | 2.02    | 11    | 2      | 9     | <2.00  | 79        |
| 25      | 2.11 | F   | 1,4   | 0       | 76     | 36      | 40      | 2.06    | 19    | 7      | 12    | 2.11   | 100       |
| 26      | 2.08 | F   | 1     | 0       | 74     | 35      | 39      | 2.05    | 18    | 7      | 11    | 2.08   | 100       |

*Note:* All scores were raw (unstandardized) scores. NSD-REP = non-arbitrary same and different relational evaluation procedure; PLS4 = Pre-School Language Scale 4th Edition; T = score for total (composite) scale; AC = auditory comprehension; EC = expressive communication; AE = age equivalent; SB5 = Stanford-Binet Intelligence Scales 5th Edition; AB = abbreviated scales; NV = nonverbal; V = verbal; IQ = intelligence quotient.

differences described here are not very large and, thus, these suggested explanations remain relatively speculative at this time. The key finding is that the NSD-REP correlated highly

with measures of language and cognition, which supports the RFT argument that relational responding is a key process underlying language and cognition.

**Table 2** Spearman's rank order correlations ( $n = 26$ ) between key assessment measures

|                | NSD-REP | PLS4 Total | PLS4-AC | PLS4-EC | SB5-ABIQ Total | SB5-V   | SB5-NV |
|----------------|---------|------------|---------|---------|----------------|---------|--------|
| NSD-REP        | -       |            |         |         |                |         |        |
| PLS4 Total     | 0.87*** | -          |         |         |                |         |        |
| PLS4-AC        | 0.88*** | 0.98***    | -       |         |                |         |        |
| PLS4-EC        | 0.84*** | 0.96***    | 0.92*** | -       |                |         |        |
| SB5-ABIQ Total | 0.82*** | 0.83***    | 0.87*** | 0.77*** | -              |         |        |
| SB5-V          | 0.79*** | 0.80***    | 0.83*** | 0.75*** | 0.97***        | -       |        |
| SB5-NV         | 0.63*** | 0.67***    | 0.72*** | 0.56**  | 0.81***        | 0.70*** | -      |

*Note:* All scores were raw (unstandardized) scores. NSD-REP = non-arbitrary same different relational evaluation procedure; PLS4 = Pre-School Language Scale 4th Edition; Total = score for total (composite) scale; AC = auditory comprehension; EC = expressive communication; SB5 = Stanford-Binet 5th Edition; ABIQ = abbreviated intelligence quotient; V = verbal; NV = nonverbal; \* =  $p < 0.05$ ; \*\* =  $p < 0.01$ ; \*\*\* =  $p < 0.001$

## Study 2

Based on the results from their initial exposure to the NSD-REP protocol in study 1, four participants were selected for training in level 1 responding (this involved selection of either the textual stimulus ‘same’ or ‘different’ given the auditory prompt “Are these the same or different?” and a pair of either identical or non-identical pictures) in the context of a controlled intervention study (i.e., a concurrent multiple-baseline design).

## Method

### *Participants and Setting*

Participants 3 (P2.4), 4 (P2.1), 6 (P2.2) and 7 (P2.3) from study 1 were employed as participants. These four participants (one girl and three boys) were chosen because (i) they had all failed level 1 of the NSD-REP; (ii) they were in roughly the same age range (i.e., 36–43 months of age,  $M = 40.8$ ,  $SD = 3.2$ ); and (iii) they shared a similar school context (i.e., they were all in the same day care facility in Limerick city). All training and assessments were conducted at a table in a corner of the general education classroom.

### *Materials*

The NSD-REP protocol was programmed in Livecode™ and presented on a Dell Latitude 2100 touchscreen laptop computer. Tangible reinforcement (e.g., stickers, erasers, sweets and colouring pages) was also used.

### *Target Behaviour and Experimental Design*

A single-subject concurrent multiple baseline design (MBD) across participants (Baer, Wolf, & Risley, 1968) was implemented. Within an MBD across participants, three or more participants are selected and the independent variable, or target intervention, is successively applied to each participant. Experimental control is established when a targeted individual’s performance improves while the other participants’ behaviour remains stable. The dependent variable or target behavior was the percentage of correct responses. Progression of treatment from one behaviour to the next was criterion-based. Participants were required to attain  $2 \times 90\%$  accuracy across 2 consecutive sessions to progress to the next phase of the intervention. On inspection of the graph, however, if there was a decreasing trend of three data points or if data stayed stationary across 3 consecutive sessions, an alternative phase was introduced. Each baseline was examined for a stable or downward trend before the next phase was introduced.

### *Procedure*

Sessions (involving either assessment or training, depending on the phase in operation for a particular participant) were conducted once a day and consisted of 20 trials. The order in which participants received their sessions on a particular day was randomized across all four. Prior to a session, the instructor asked the participant concerned if they would like to play the same and different game on the computer. If the participant agreed, he or she was brought individually to the teacher’s desk in the corner of the room.

**Phase I Baseline** During this phase, participants were exposed to level 1 of the NSD-REP as described in study 1 and involving one (randomized) stimulus set. There was no feedback given for correct or incorrect responses during baseline sessions.

**Phase II Training** After three baseline sessions, participant 1 commenced training. This phase differed from baseline in that auditory feedback (i.e., ‘yes’ for correct and ‘no’ for incorrect) was provided after each response. In addition, the stimulus set used during baseline was not employed during this phase and instead a number of novel stimulus sets were used. During this phase, a stimulus set was used for two consecutive sessions before a novel stimulus set was introduced. This switching, which was independent of accuracy on a particular stimulus set, was effected to try to bring responding more under the influence of the non-arbitrary same and different relations involved than features of particular stimulus sets. The first participant (P2.1) commenced training after three baseline sessions. Each subsequent participant’s intervention started after the previous participant had achieved criterion and a minimum of three sessions after the previous participant’s intervention had started.

**Phase III Baseline** When participants reached criterion in the training phase (i.e., 90 % correct across two consecutive sessions), they were re-tested with the stimulus set from phase I. This occurred one week post-intervention. Identical to the initial baseline, no feedback was provided during this phase.

**Phase IV Follow-Up** During this phase, generalised responding to non-arbitrary same and different relations was assessed by introducing a novel stimulus set and providing no feedback during testing. This was carried out two weeks subsequent to the end of training.

## Results and Discussion

P2.1 took a total of 6 sessions from start to finish to complete participation, with a mean of 1 m 32 s per session; P2.2 took a



total of 8 sessions, with a mean of 1 m 30 s per session; P2.3 took a total of 7 sessions, with a mean of 2 m 01 s per session; and P2.4 took a total of 9 sessions, with a mean of 2 m 25 s per session.

Figure 2 displays the effects of training on the percentage of correct responses across four phases (i.e., baseline, training, baseline and follow-up) for the four participants. Training produced an increase in the percentage of correct responses for all participants. P2.1 and P2.3 demonstrated criterion level responding within two sessions of intervention and ultimately only required three training sessions to meet the criterion fully. They maintained these high scores when the baseline stimulus set was re-administered 1 week later and sustained them also in a test with a novel stimulus set administered 2 weeks after training had finished. P2.2 and P2.4 completed training in four and five sessions, respectively, and their scores for baseline and follow-up were comparable to those of P2.1 and P2.3.

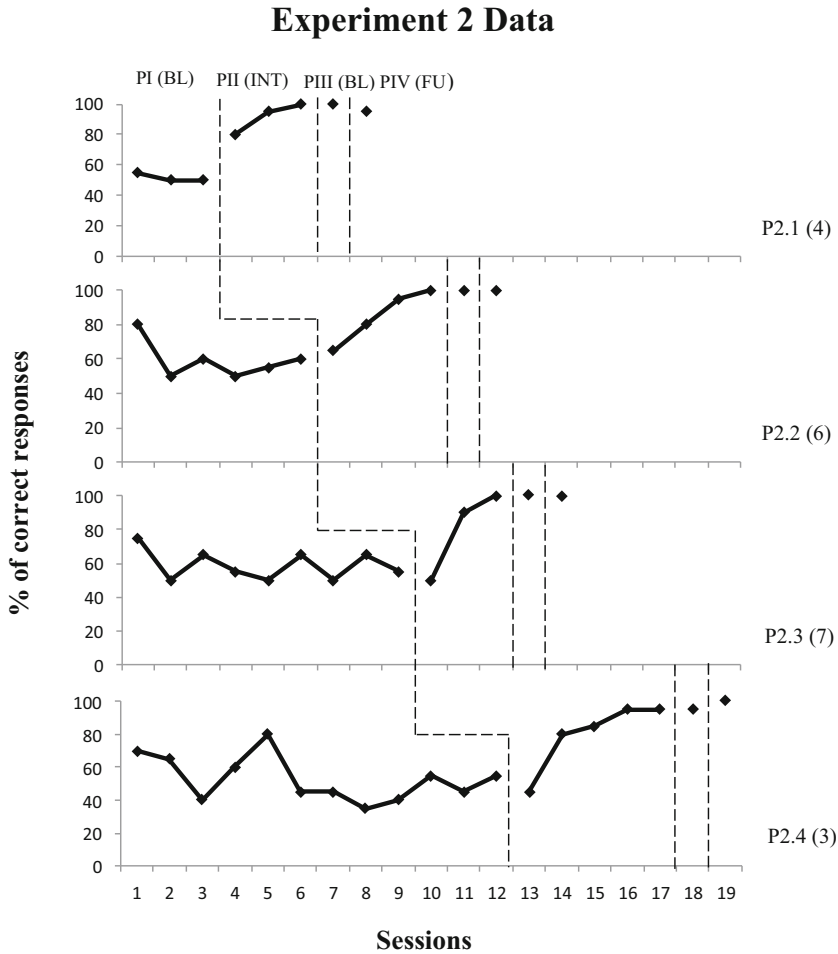
The results of this second study demonstrate the efficacy of the NSD-REP format for training simple REP-type relational responding in young children. Specifi-

cally, this study successfully used a multiple baseline design to establish correct responding to level 1 of the protocol, which required responding to ‘same’ and ‘different’ as textual stimuli in response to identical and non-identical pictorial stimuli, respectively. Given this success with level 1 of the protocol, the aim of the next study was to examine whether a similar approach might be used to establish correct responding to level 2, which required more complex responding involving selection of ‘yes’ and ‘no’ in response to questions concerning the description of same/different non-arbitrary relations.

Study 3

To extend the results of the previous study, study 3 aimed to investigate the possibility of teaching participants to successfully show NSD-REP level 2 responding, which involved responding ‘yes’ or ‘no’ to non-arbitrary same and different relations. Once again, a multiple baseline design was employed.

**Fig. 2** Percentage of correct responses for level 1 of the NSD-REP during PI baseline (BL), PII intervention (INT), PIII baseline (BL) and PIV follow-up (FU), across each participant. Intersection lines indicate progression to the next phase of the study. Numbers in brackets after participant numbers indicate number of that participant in study 1



## Method

### *Participants and Setting*

Participants 1 (P3.4), 2 (P3.2), 4 (P3.5), 7 (P3.7), 12 (P3.3), 14 (P3.1), 15 (P3.6) and 21 (P3.8) from study 1 were employed again as participants in the present study. These 8 children, including 4 girls and 4 boys, ranged in age from 37–59 months ( $M = 49.5$ ,  $SD = 8.0$ ). They came from two separate facilities in Limerick city including a city-based day care facility and a Montessori school (for a description of both, see study 1).

### *Materials*

As in study 2, the NSD-REP protocol was administered using a touch screen Dell Latitude 2100 computer. Tangible reinforcers (e.g., stickers, erasers, sweets and colouring pages) were also employed.

### *Target Behaviour and Experimental Design*

This study was implemented across two educational settings. A non-concurrent multiple baseline across participants design was used. The target behaviour was correct NSD-REP level 2 responding (i.e., choosing either ‘yes’ or ‘no’ as appropriate in response to trials involving the juxtaposition of non-arbitrary same or different and an auditory cue of ‘same’ or ‘different’). The pass/progression criteria were the same as for study 2.

### *Procedure*

A maximum of one session (of 20 trials) was conducted per day per participant (involving either assessment or training, depending on the phase in operation for a particular participant). Apart from this constraint, sessions were conducted as frequently as possible across the two facilities involved. Prior to a session, the instructor asked the participant concerned if they would like to play the same and different game on the computer. If the participant agreed, he or she was assessed and/or trained individually in a quiet location where the session could be conducted undisturbed.

**Phase I Baseline** During this phase, participants were exposed to NSD-REP level 2 as described in study 1 with one (randomized) stimulus set (i.e., involving 20 pairs of visual stimuli). There was no feedback for correct or incorrect responses during baseline sessions.

**Phase II Training** After three baseline sessions, P3.1 and P3.2 commenced training. This phase differed from baseline in that (auditory and visual) feedback were provided for correct and incorrect responding. After a correct response, the computer presented the auditory stimulus ‘correct, well done’

accompanied by a visual display consisting of a grid of colored stimuli including stars, small animals and shapes. After this, the screen cleared for 0.5 seconds before the start of the next trial. For incorrect responses, the computer presented the auditory stimulus ‘wrong’ with no visual accompaniment and the program automatically stopped to allow the teacher to provide feedback. She did this by repeating the auditory (question) prompt and providing a further vocal and gestural prompt to highlight the correct answer. During the training phase, a stimulus set was used for two consecutive sessions before a novel stimulus set was introduced. P3.3 and P3.4 started training after 6 baseline sessions; P3.5 and P3.6 started training after 9 baseline sessions; and P3.7 and P3.8 began training after 12 baseline sessions.

**Phase III Baseline** After participants reached criterion in phase II (i.e., 90 % correct in two consecutive sessions), they were re-tested with the stimulus set from phase I. This occurred 1 week post-intervention. As in the (phase I) baseline, no feedback was provided.

**Phase IV Follow-Up** This phase assessed for generalization and maintenance of NSD-REP level 2 responding by assessing performance on this level with a novel stimulus set and no feedback. This assessment was carried out two weeks after the end of training.

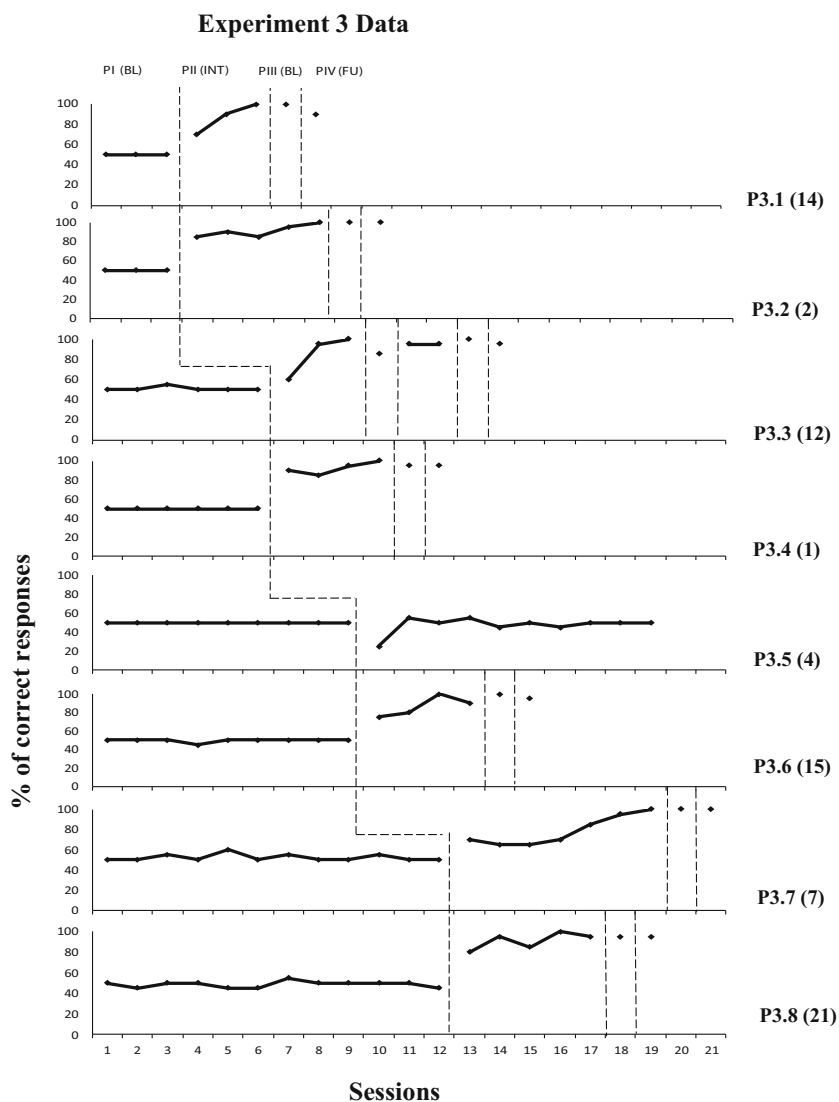
**Post Assessment** Following phase IV, participants from both studies 2 and 3 were re-tested using the SB-5, PLS-4 and NSD-REP.

## Results and Discussion

P3.1 took a total of 9 sessions, with a mean of 1 m 32 s per session; P3.2 took a total of 11 sessions, with a mean of 1 m 56 s per session; P3.3 took a total of 9 sessions, with a mean of 1 m 38 s per session; P3.4 took a total of 10 sessions, with a mean of 1 m 27 s per session; P3.5 took a total of 12 sessions, with a mean of 1 m 32 s per session; P3.6 took a total of 9 sessions, with a mean of 1 m 24 s per session; P3.7 took a total of 9 sessions, with a mean of 2 m 01 s per session; and P3.8 took a total of 11 sessions, with a mean of 1 m 30 s per session.

Figure 3 shows the results of the concurrent multiple baseline design procedure employed in the present study to show the effect of NSD-REP level 2 (yes and no) training. As in study 2, there were four phases (i.e., P I Baseline; P II Training; P III Baseline; and P IV Follow-up). As can be seen, all participants maintained a relatively stable performance during phase I (baseline). In fact, this data seems extremely stable, hovering around 50 % across sessions. This stability can be traced to the fact that many participants tended to choose the leftmost (i.e., ‘yes’) button consistently across trials. Phase

**Fig. 3** Percentage of correct responses for level 2 of the NSD-REP during phase I baseline (BL), phase II intervention (INT), phase III baseline (BL) and phase IV follow-up (FU), in experiment 3, for each of the eight participants involved. Intersection lines indicate progression to the next phase of the study. Numbers in brackets after participant numbers indicate number of that participant in study 1



II (intervention) resulted in an increase in the percentage of correct responses in the case of all but one of the participants. P3.1, P3.2, P3.4, P3.6, P3.7 and P3.8 demonstrated an immediate increase in the percentage of correct responding and quickly reached criterion. Furthermore, they maintained criterion level responding in phase III (return to baseline) and phase IV (follow-up). P3.3 met criterion in phase II within four sessions but failed to maintain criterion responding in phase III (return to baseline). Hence, phase II was repeated. After reaching criterion in two sessions, P3.3 was then re-exposed to phase III (return to baseline) and passed and then passed also in phase IV (follow-up).

P3.5 was exposed to phase II after 9 exposures to phase I; however, this participant failed to reach criterion after 10 sessions of training and, thus, in accordance with pre-defined criteria, their participation was ended. It should be noted that this participant had previously taken part in study 2, which

trained level 1 responding and their responding on level 2 seemed to be influenced by the training received in that previous study in that they vocalised 'same' and 'different' when responding to the Yes and No button stimuli.

The main purpose of Studies 2 and 3 was to investigate the efficacy of multiple exemplar training (MET) for improving the performance of young children on various levels of the NSD-REP protocol as a methodology for training relational responding. Overall, the results of these studies support the conclusion that young children aged between three and four can be taught to respond on the NSD-REP. Given the success with this age group, one question that arises is whether an even younger cohort (in the 2.5 – 3 year range) with a potentially less developed repertoire (as measured using similar standardised testing procedures) might also be trained. This was the focus of study 4.

## Study 4

Study 4 aimed to teach NSD-REP level 1 responding (i.e., non-arbitrary same and different relations) to two young children in the 2.5 – 3 year range.

### Method

#### *Participants and Setting*

Participants 23 (P4.2) and 25 (P4.1) from study 1, both of whom had failed level 1 of the NSD-REP, participated in this study. P4.1 (female) was 35 months old while P4.2 (female) was 33 months old. Both participants were recruited from a daycare facility in Limerick city. (see study 1). As may be seen in Table 1, both P4.1 and P4.2 had lower scores than any of the participants in study 2 (which had also focused on training level 1) on a number of key measures including PLS-4, PLS-4 age equivalent, SB5-ABIQ and SB5-ABIQ verbal.

#### *Materials*

As in study 2, the NSD-REP protocol was presented on a touch screen Dell Latitude 2100 computer. In addition, a number of stimulus sets each made up of ten picture pairs (including pictures of animals, shapes, common objects and cartoon characters) were used during table top-based stages of assessment and training. Tangible reinforcers (e.g., stickers, erasers, sweets and colouring pages) were also employed.

#### *Target Behavior and Experimental Design*

The target behavior was similar to study 2; however, the core intervention used a table top rather than a touch-screen computer-based methodology. In its essence, the design could be considered an ABA design conducted for each of two participants, where the A phase was performance on a computer-presented task while the B phase involved remediation using a table top format. At the same time, however, both the A and B phases involved sub-phases of testing and training and, thus, in that sense, the design is more complex than a simple ABA design. The same pass criteria as for previous studies were employed.

#### *Procedure*

**Phase A – NSD-REP Level 1 Computer-Based Assessment & Training** The first section of this phase involved assessing participants on level 1 of the NSD-REP protocol without feedback. If participants failed criterion, they were moved into NSD-REP level 1 training. This was the same as the previous part except it included feedback. If participants passed training, they

would be moved into the previous (assessment) part. If they failed training, they were moved to table top assessment and training.

**Phase B – Table Top Assessment & Training.** This phase included a number of sub-phases.

**Table Top Sub-Phase 1** During this sub-phase, participants were exposed to a quasi-random sequence of table top-presented same and different trials. During each trial, two pictures (either identical or non-identical) were laid face up on the table in front of the participant and the researcher then presented an auditory S<sup>D</sup> (i.e., ‘Are these the same or different?’). A correct response involved saying ‘same’ if the pictures were identical and ‘different’ if not. No feedback was provided after a response; instead, the pictures were collected and those for the next trial presented. A total of 20 trials (10 ‘same’ and 10 ‘different’) was presented. If participants passed, then they would be re-exposed to the NSD-REP level 1 training. If not, they went to sub-phase 2 of the table top protocol.

**Table Top Sub-Phase 2** This was essentially the same as sub-phase 1 except that participants received feedback (either ‘correct’ or ‘wrong’). It involved two sub-sub-phases: (i) where trials were organised in alternating three-trial sequences involving either all ‘same’ or all ‘different’ trials; and (ii) where trials were presented quasi-randomly in pairs involving just a single ‘same’ and a single ‘different’ trial. If participants passed sub-phase 2 then they would be given sub-phase 1 again. If they failed, they would be moved to sub-phase 3.

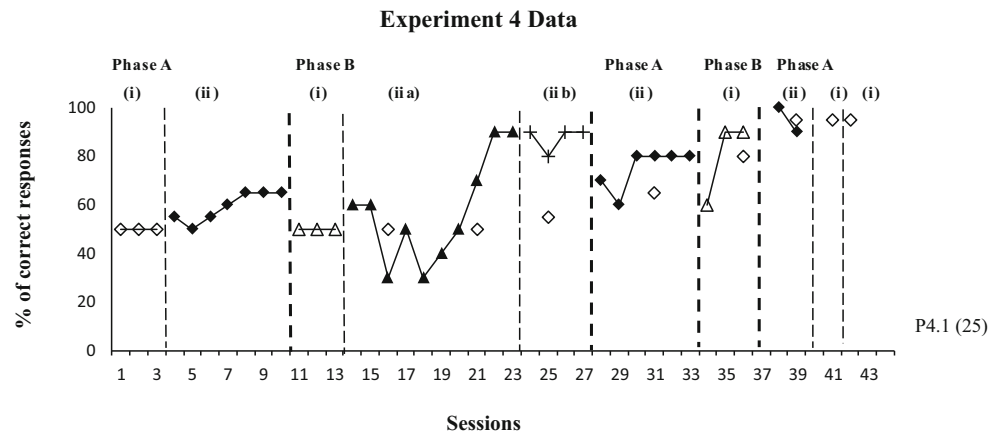
**Table Top Sub-Phase 3** This sub-phase involved using an all visual conditional discrimination format with a sample picture which was given to the participant and two comparison pictures, one identical to the sample and one non-identical, laid out in front of the participant. On each trial, the researcher asked the participant to either ‘Find a picture that is the same’ as the sample or ‘Find a picture that is different’ from the sample. This sub-phase involved three sub-sub-phases: (i) same trials only; (ii) different trials only; and (iii) a quasi-random mix of same and different trials.

After successfully completing phase B, participants were re-exposed to phase A.

### Results and Discussion

Figure 4 displays the data for each of the two participants in study 4, with the upper and lower panels showing the data for P4.1 and P4.2 respectively. P4.1 took a total of 42 sessions with a mean of 3 m 23 s per session; P4.2 took a total of 76 sessions with a mean of 3 m 33 s per session. P4.1 failed both testing and training in the A (computer-based) phase and, thus, was exposed to the B (table top) phase. This participant appeared to progress relatively quickly through the table top procedures but

**Fig. 4** *Upper panel:* Percentage of correct responses for level 1 of the NSD-REP during various phases (see legend) for participant 4.1. *Lower panel:* Percentage of correct responses for level 1 of the NSD-REP during various phases (see legend) for participant 4.2



did not show immediate improvement on the phase A probes. After re-exposure to additional phase B training, however, this participant showed improvement in phase A probes and then passed both training and testing of phase A. P4.2 failed both testing and training in the A (computer-based) phase and, thus, was exposed to the B (table top) phase. P4.2 required greater levels of exposure to B sub-phases than the first participant. Eventually, however this participant met training criteria for these phases, after which she went on to successfully complete both training and testing in phase A. Both participants maintained their scores when a follow-up (generalisation) stimulus set (without feedback) was administered one week later.

## General Discussion

Several previous studies have demonstrated the potential of the REP as a means of assessing and training relational responding. Most of this work has been conducted with adults or at least school age children (e.g., Cassidy, Roche & Hayes, 2011; Cassidy, Roche, Colbert, Stewart & Grey 2016; Hayes & Stewart, 2016). The current series of studies aimed to extend this work by investigating the utility of the REP as a means of assessing and training relational responding in very young (i.e., pre-school) children. Typically developing pre-school children ranging from 2 to 5 years of age were assessed and trained on an REP type protocol, the NSD-REP, whose core focus was non-arbitrary same and difference relations. In study 1, a correlational study involving a sample size of 26, NSD-REP performance was shown to correlate highly with performance on mainstream measures of language and cognition (i.e., the Stanford Binet 5th Edition [SB5] and the Pre-School Language Scale 4th Edition [PLS-4]). In studies 2–4, groups of children not meeting criterion on one or other of the levels of the NSD-REP were successfully trained to pass on that level.

When examining the correlations amongst the NSD-REP, the PLS-4 and the SB5-ABIQ in study 1, it was found that the highest correlation was between the NSD-REP and the PLS-4;

and that both these measures correlated at a lower level but still strongly and comparably with the SB5-ABIQ. The close correlation of the NSD-REP with both other assessments reflects the RFT conception that contextually controlled relational responding is the core process involved in language and cognition. The stronger correlation with the measure of language than with the measure of cognition may reflect a closer connection between relational responding and focused language tasks than between relational responding and a broader range of tasks. It was argued that the pattern of correlations between the NSD-REP and the SB5-ABIQ subscales supports this suggestion; the correlation with the verbal subscale was higher than with the nonverbal subscale. The correlations between the NSD-REP and the PLS-4 subscales were also noted. In this case, it was seen that the NSD-REP correlated more highly with the auditory comprehension subscale than with the expressive communication subscale and indeed it correlated more highly with the former than with the PLS-4 itself. We pointed out that this might be a result of the inclusion of auditory as well as visual stimuli as part of the trials on all three stages of the NSD-REP and the fact that touchscreen rather than spoken responses were required. While these suggestions remain as yet speculative, the broad pattern of correlations supports the RFT argument that relational responding is a key process underlying language and cognition.

The finding that relational responding correlates well with performance on measures of language and cognition is not new. In fact, this is now a relatively well-established empirical finding based on work with both adults and children and involving individuals with developmental delay as well as typically developing populations (e.g., Moran, Stewart, McElwee, & Ming, 2010; Moran, Stewart, McElwee, & Ming, 2014; O'Toole & Barnes-Holmes, 2009; see Stewart and Roche, 2013, for a review of previous research). However, one feature of the current work which may be unique is the relative ease of administration and, thus, practical utility of the protocol used to gauge level of relational responding. The correlations with mainstream measures were



high, suggesting that the protocol is indeed a useful measure of relational responding as a core aspect of intellectual performance. At the same time, the assessment was brief (e.g., average assessment time of less than 6 minutes) and unproblematic (all children could be tested) as a means of measuring relational performance even in very young children. As such, this protocol or something similar may be a useful gauge in future studies, behavioral or otherwise, not just of relational responding but of intellectual performance more generally.

Apart from constituting a successful means of assessing relational responding, the current research also illustrates the utility of the NSD-REP as a means of training this repertoire. As reported, over the course of a number of experiments, the protocol was employed successfully to train patterns of relational responding in very young children, including a number of children under the age of 3. This suggests that the REP is a potentially useful method for training relational responding in very young populations; furthermore, it recommends in particular the NSD-REP as a useful exemplar of the REP.

Hence, the current research showed improvement in relational responding based on REP training. One way in which future work might advance from this is to examine the possible effects of such training on intellectual performance. Previous research (Cassidy, Roche, & Hayes, 2011; Cassidy, Roche, Colbert, Stewart, & Grey, 2016; Hayes & Stewart, 2016) has found that relational training can boost children’s performance on standardized tests of language and cognition. Performance on tests such as these in turn is widely interpreted as an indicator of intellectual (linguistic and cognitive) potential; in other words, as a predictor of performance in academic and career contexts (e.g., Fergusson, Horwood, & Ridder, 2005). The children trained in previous studies have been in middle-to-late childhood. However, the current study has shown the possibility of REP training in relatively young children. Given the gains in intellectual potential that older children have shown based on relational training, it seems important and timely to investigate whether similar or perhaps even increased benefits might accrue to children given such training at a younger age. Future work might investigate this possibility by giving children of similar age to those in the current study systematic training using the REP over a substantial period of time (e.g., 6 months or longer) and with suitable adaptations for the age group involved such as were employed in the current study (e.g., focusing on primarily non-arbitrary relations, briefer duration of individual training sessions, higher levels of teacher supervision and higher levels of reinforcement). A training group might be compared with an active comparison condition and/or ‘no treatment’ controls in a randomised controlled trial for improvements in performance on standardised ability tests. Given the potential benefits of starting intellectual training at a younger age, such work might have substantive potential.

Collectively, these findings add to a growing literature that supports the use of a relational frame theory approach and more specifically, an REP-based approach to assessing and training generalized relational responding.

Compliance with Ethical Standards

**Conflict of Interest** The first author declares that she has no conflict of interest. The second author declares that he has no conflict of interest. The third author declares that he has no conflict of interest.

**Ethical Approval** All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

**Informed Consent** Informed consent was obtained from all individual participants included in the study.

Appendix 1

**Table 3** Vocal response targets and antecedent stimuli presented under tact conditions for a representative participant

| Vocal response targets | Antecedent verbal stimulus | Antecedent nonverbal stimulus |
|------------------------|----------------------------|-------------------------------|
| Tact “yes”             | “Is this a cup?”           | Picture of a cup              |
|                        | “Is this a boat?”          | Picture of a boat             |
|                        | “Is this a bear?”          | Picture of a bear             |
|                        | “Is this a car?”           | Picture of a car              |
|                        | “Is this a cow?”           | Picture of a cow              |
| Tact “no”              | “Is this a shoe?”          | Picture of a boat             |
|                        | “Is this a pen?”           | Picture of a bear             |
|                        | “Is this a ball?”          | Picture of a cup              |
|                        | “Is this a duck?”          | Picture of a car              |
|                        | “Is this a house?”         | Picture of a cow              |

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